INCH-POUND

MIL-C-24643/19C
14 March 1994
SUPERSEDING
MIL-C-24643/19B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, TYPE LS6SGU (INCLUDING VARIATION LS6SGA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (watertight with circuit integrity)

First - Copper conductor, uncoated, (see table I for size)
Second - Silicone rubber - glass tape insulation (see table I for diameter). Letter identification code applied by method 5 (two conductors each A, B and C).
* Third - The six conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly. Cabling sequence shall be A, B and C.
Fourth - A binder or combination binder and barrier.

Construction variant for type LS6SGU

* Fifth - Cross-linked polyolefin jacket. (see table I for thickness).

Construction variant for type LS6SGA

* Fifth - Cross-linked polyolefin jacket. (see table I for thickness).
* Sixth - Braided metal armor.

AMSC N/A FSC 6145
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TABLE I. Details.

Military 1/ part no. M24643/19	Size 2/	Conductor size AWG	Type's overall diameter (inches)				Dia over insulation min avg (in.)	Conductor resistance per 1000 feet at 25 deg C max (ohms)	Jacket thickness min avg (in.)	Insulation resistance (megohms)	Cold bending mandrel dia (max) (in.)	Accelerated service loading ampere
			LS6SGU		LS6SGA							
			min	max	min	max						
-01UN	100	0 (class D)		1.600		1.650	0.453	0.106	0.060	160	20	475
-02UN	125	00 (class D)		1.790		1.840	.507	.0850	.060	150	22	520
-03UN	150	000 (class D)		1.960		2.010	.557	.0674	.060	135	24	570
-04UN	200	0000 (class D)		2.200		2.250	.634	.053	.075	125	27	650

1/ Unarmored. When armored option is required, see MIL-C-24643 for configuration of part number.

2/ Type LS6SGU and LS6SGA

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	(see table I)
Voltage withstand - volts, root mean square, minimum	5000
Insulation resistance - megohms/1000 feet, minimum	
* Conductor to conductor	(see table I)
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for limits of water leakage	No failure
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Gas Flame - 1 hour	No failure
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Cold bending, cable - (see table I for mandrel diameter)	No damage
Physicals (unaged)	
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
* Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - cycles, minimum	125
* Cable sealant removability	No failure
Armor (type LS6SGA only) - conformance to material construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Cold working (minus 20 ± 2°C	No damage
Gas Flame (3 hours)	No failure
Accelerated service - sizes 50 through 800 (see table I for load current)	No failure
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
* Insulation	35
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

All sizes 800 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH

Review Activities:
Army - AV, CR, ER, ME, AR, AL
Navy - EC, CG
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